

# Artavazd Maranjyan

✉ [arto.maranjyan@gmail.com](mailto:arto.maranjyan@gmail.com)

🌐 [artomaranjyan.github.io](https://artomaranjyan.github.io)

🔍 [Google Scholar](#) [LinkedIn](#)

## Current Position

2026–present **Postdoctoral Researcher and EPFL AI Center and Swiss AI Initiative Postdoctoral Fellow**, EPFL, Lausanne, Switzerland  
Jointly hosted by **Volkan Cevher**'s LIONS group and **Martin Jaggi**'s MLO group.  
Developing theoretically grounded methods to make large-scale AI training faster and more resource-efficient, with emphasis on distributed and asynchronous optimization, communication efficiency, and model parallelism.

## Education

2023–2025 **Ph.D. in Computer Science**, KAUST, Thuwal, Saudi Arabia; **Advisor**: Peter Richtárik  
**Thesis**: [First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data](#)

2021–2023 **M.Sc. in Applied Statistics and Data Science**, Yerevan State University, Yerevan, Armenia  
**Thesis**: [On local training methods](#); **Co-supervisors**: Peter Richtárik, Mher Safaryan

2017–2021 **B.Sc. in Informatics and Applied Mathematics**, Yerevan State University, Yerevan, Armenia  
**Thesis**: [On the Convergence of Series in Classical Systems](#); **Supervisor**: Martin Grigoryan

## Research Experience

03/2025 **Visiting Researcher, hosted by Yi-Shuai Niu**, BIMSA, Beijing, China

- Collaborated on server-assisted federated learning.
- Delivered invited seminars at Peking University (PKU), Beihang University (BUAA), and BIMSA.

04/23–08/23 **Researcher, Martin Grigoryan's group**, Yerevan State University, Yerevan, Armenia

- Investigated universal functions for the Vilenkin and Haar systems across different function spaces.

06/22–01/23 **Research Intern, Peter Richtárik's group**, KAUST, Thuwal, Saudi Arabia

- Contributed to GradSkip (TMLR), a communication-efficient local optimization method with improved complexity.

01/22–08/23 **Machine Learning Researcher, YerevaNN**, Yerevan, Armenia

- Researched optimization methods for federated learning.

## Industry Experience

07/21–06/22 **Co-Founder, OnePick**, Yerevan, Armenia

- Built an MVP that helped content creators and marketers select photos matching their text descriptions.
- Advanced through [FAST's InVent 2.0](#): team formation, ideation, product development, and pitching.
- Secured initial funding and continued development beyond the program.

07/21–09/21 **Backend Developer, EXALT Technologies Ltd**, Yerevan, Armenia

- Contributed to backend development for a Nutanix project.

06/21–07/21 **Machine Learning Research Engineer, FAST**, Yerevan, Armenia

- Developed machine-learning models for fraud detection.
- Built machine-learning and statistical models for data-driven forecasting.

09/19–01/21 **Software Engineer in Test, Picsart**, Yerevan, Armenia

- Developed test-automation solutions across several mobile and web applications.
- Collaborated across engineering, QA, product, and business teams to diagnose and resolve product issues.
- Proposed improvements to DevOps processes and Jenkins CI pipelines.

## Awards

06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowship**, EPFL  
Competitive fellowship supporting independent research, with an additional travel fund (CHF 5,000 annually)

05/2025 **CEMSE Dean's List Award**, KAUST  
Awarded for excellent academic and research performance (USD 2,500)

09/2023 **Dean's Award**, KAUST  
Merit-based award for selected incoming KAUST students (USD 18,000 over three years)

05/2021 **Outstanding Final Project Award**, Yerevan State University  
Recognized for an outstanding bachelor's thesis (one of 6 recipients among 250+ students)

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## Publications

15. “Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity.” A. Mahran, A. Maranjyan, P. Richtárik. **NeurIPS** 2026, [arXiv:2605.13434](#); [poster].
14. “LOSCAR-SGD: Local SGD with Communication-Computation Overlap and Delay-Corrected Sparse Model Averaging.” Y. Maziane, A. Mahran, A. Maranjyan, P. Richtárik. **NeurIPS** 2026, [arXiv:2605.20866](#).
13. “Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method.” A. Sadiev, A. Maranjyan, I. Ilin, P. Richtárik. [arXiv:2605.18174](#), 2026.
12. “Rennala MVR: Improved Time Complexity for Parallel Stochastic Optimization via Momentum-Based Variance Reduction.” Z. Tovmasyan, A. Maranjyan, P. Richtárik. [arXiv:2605.08871](#), 2026.
11. “Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity.” A. Maranjyan, P. Richtárik. **ICLR** 2026, [arXiv:2509.22860](#); [poster] [slides].
10. “BiCoLoR: Communication-Efficient Optimization with Bidirectional Compression and Local Training.” L. Condat, A. Maranjyan, P. Richtárik. [arXiv:2601.12400](#), 2026.
9. “ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning.” A. Maranjyan, E. M. Saad, P. Richtárik, F. Orabona. **ICML** 2025, [arXiv:2502.00775](#); [poster].
8. “Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity.” A. Maranjyan, A. Tyurin, P. Richtárik. **ICML** 2025, [arXiv:2501.16168](#); [poster] [slides] [video].
7. “MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times.” A. Maranjyan, O. S. Omar, P. Richtárik. **UAI** 2025; **OPT** 2024, **Oral (top 5%)**, [arXiv:2410.04285](#); [poster] [slides] [video].
6. “GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity.” A. Maranjyan, M. Safaryan, P. Richtárik. **TMLR** 2025, [arXiv:2210.16402](#); [poster] [slides] [video].
5. “LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression.” L. Condat, A. Maranjyan, P. Richtárik. **ICLR** 2025, **Spotlight (top 5.1%)**, [arXiv:2403.04348](#); [poster].
4. “Differentially Private Random Block Coordinate Descent.” A. Maranjyan, A. Sadiev, P. Richtárik. **OPT** 2024, [arXiv:2412.17054](#); [poster].
3. “Menshov-type theorem for divergence sets of sequences of localized operators.” M. Grigoryan, A. Kamont, A. Maranjyan. *Journal of Contemporary Mathematical Analysis*, vol. 58, no. 2, pp. 81–92, 2023.
2. “On the divergence of Fourier series in the general Haar system.” M. Grigoryan, A. Maranjyan. *Armenian Journal of Mathematics*, vol. 13, pp. 1–10, 2021.
1. “On the unconditional convergence of Faber-Schauder series in  $L^1$ .” T. Grigoryan, A. Maranjyan. *Proceedings of the YSU A: Physical and Mathematical Sciences*, vol. 55, no. 1 (254), pp. 12–19, 2021.

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## Reviewing

TMLR (2024–present); NeurIPS and ICML (2026); ICLR (2025); SIMODS and JMLR (2024).

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## Selected Talks and Posters

- 08/2026 Poster, **Swiss Optimization Symposium**, Ascona, Switzerland [poster]
- 04/2026 **Invited talk**, **Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)**, Rio de Janeiro, Brazil [video] [slides]
- 07/2025 **Talk and poster**, **35<sup>o</sup> Colóquio Brasileiro de Matemática**, IMPA, Rio de Janeiro, Brazil [slides] [poster]
- 04/2025 **Invited talk**, **Federated Learning One World Seminar (FLOW)**, Online [video] [slides]
- 03/2025 **Invited talk**, **Academic Report of the School of Mathematical Sciences**, Beihang University (BUAA), Beijing, China, hosted by Jiaxin Xie [slides]
- 03/2025 **Invited talk**, **Optimization Seminar**, BIMS, Beijing, China, hosted by Yi-Shuai Niu [slides]
- 03/2025 **Invited talk**, **Seminar**, Peking University, Beijing, China, hosted by Kun Yuan [slides]
- 12/2024 **Talk (oral, top 5%)** **Optimization for Machine Learning (NeurIPS 2024 Workshop)**, Vancouver, Canada [video]
- 11/2024 **Invited talk**, **MLR Weekly Seminar**, Apple, Online, hosted by Samy Bengio [slides]
- 03/2024 Poster, **The Machine Learning Summer School in Okinawa 2024**, OIST, Okinawa, Japan [poster]
- 10/2023 **Invited talk**, **Algorithms & Computationally Intensive Inference seminars**, University of Warwick, Coventry, England [slides]
- 12/2022 **Invited talk**, **Federated Learning One World Seminar (FLOW)**, Online [video]